

Adrenalin Enterprise Ticketing System

Schema Reference — v7 (Normalized Lookups Addendum)

PostgreSQL 15+ · 48 tables across 12 schemas · Generated May 2026

This document describes every schema, table, and trigger defined in `schema_creation-v2.sql`. It covers table purpose, column definitions (name, type, constraints, description), and the reusable trigger functions applied across the database.

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Triggers	2 reusable trigger functions (set_updated_at, generate_ticket_number)

Schema: lookup

Master reference / config tables used across all domains.

Table	Description
<code>lookup.customer_tiers</code>	M1-M4 customer tiers. <code>priority_bump</code> elevates ticket priority at creation.
<code>lookup.geo_regions</code>	Region-specific business hours and timezone. SLA clock runs only within these windows.
<code>lookup.product_versions</code>	Named product release lines. Soft-deleted versions remain on historical tickets.
<code>lookup.modules</code>	Top-level product functional areas. Drives graph scope resolution and SLA routing.
<code>lookup.sub_modules</code>	Fine-grained ticket classification. <code>requires_dual_confirm</code> triggers GRAPH-006; <code>requires_compliance_review</code> triggers GRAPH-005.

lookup.customer_tiers

M1-M4 customer tiers. `priority_bump` elevates ticket priority at creation.

Column	Data Type	Constraints	Description
<code>code</code>	<code>varchar(10)</code>	PK	Tier code e.g. M1, M2
<code>label</code>	<code>varchar(80)</code>	NOT NULL	Human-readable name
<code>description</code>	<code>text</code>		Optional description
<code>priority_bump</code>	<code>integer</code>	NOT NULL DEFAULT 0	SLA priority elevation (0–2)
<code>is_active</code>	<code>boolean</code>	NOT NULL DEFAULT TRUE	Soft-disable flag
<code>created_at</code>	<code>timestampz</code>	NOT NULL DEFAULT NOW()	Creation timestamp

lookup.geo_regions

Region-specific business hours and timezone. SLA clock runs only within these windows.

Column	Data Type	Constraints	Description
<code>code</code>	<code>varchar(20)</code>	PK	Region code e.g. SASIA, EUR
<code>label</code>	<code>varchar(80)</code>	NOT NULL	Display name
<code>timezone</code>	<code>varchar(60)</code>	NOT NULL	IANA timezone e.g. Asia/Kolkata
<code>business_start</code>	<code>time</code>	NOT NULL	Business hours start
<code>business_end</code>	<code>time</code>	NOT NULL	Business hours end
<code>working_days</code>	<code>varchar(20)</code>	NOT NULL DEFAULT Mon-Fri	Working day range
<code>is_active</code>	<code>boolean</code>	NOT NULL DEFAULT TRUE	Active flag
<code>created_at</code>	<code>timestampz</code>	NOT NULL DEFAULT NOW()	Creation timestamp

lookup.product_versions

Named product release lines. Soft-deleted versions remain on historical tickets.

Column	Data Type	Constraints	Description
<code>id</code>	<code>uuid</code>	PK DEFAULT <code>gen_random_uuid()</code>	Surrogate key
<code>code</code>	<code>varchar(50)</code>	UNIQUE NOT NULL	Lowercase version code

Column	Data Type	Constraints	Description
label	varchar(150)	NOT NULL	Display label
release_date	date		Release date
is_legacy	boolean	NOT NULL DEFAULT FALSE	Legacy flag
is_active	boolean	NOT NULL DEFAULT TRUE	Active flag
is_deleted	boolean	NOT NULL DEFAULT FALSE	Soft-delete flag
created_by	uuid	FK auth.users	Creator user
updated_by	uuid	FK auth.users	Last updater
created_at	timestamptz	NOT NULL DEFAULT NOW()	Creation timestamp
updated_at	timestamptz	NOT NULL DEFAULT NOW()	Last update timestamp

lookup.modules

Top-level product functional areas. Drives graph scope resolution and SLA routing.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
code	varchar(80)	UNIQUE NOT NULL	Lowercase module code
label	varchar(150)	NOT NULL	Display name
description	text		Optional description
department	varchar(80)		Support Delivery PMO
is_mandatory_for_closure	boolean	NOT NULL DEFAULT TRUE	Must be set before ticket close
is_active	boolean	NOT NULL DEFAULT TRUE	Active flag
is_deleted	boolean	NOT NULL DEFAULT FALSE	Soft-delete
created_by	uuid	FK auth.users	Creator
updated_by	uuid	FK auth.users	Updater
created_at	timestamptz	NOT NULL DEFAULT NOW()	Creation time
updated_at	timestamptz	NOT NULL DEFAULT NOW()	Update time

lookup.sub_modules

Fine-grained ticket classification. requires_dual_confirm triggers GRAPH-006; requires_compliance_review triggers GRAPH-005.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
module_id	uuid	NOT NULL FK lookup.modules	Parent module
code	varchar(100)	UNIQUE NOT NULL	Lowercase sub-module code
label	varchar(200)	NOT NULL	Display name
description	text		Optional description
requires_dual_confirm	boolean	NOT NULL DEFAULT FALSE	Triggers GRAPH-006
requires_compliance_review	boolean	NOT NULL DEFAULT FALSE	Triggers GRAPH-005
is_active	boolean	NOT NULL DEFAULT TRUE	Active flag
is_deleted	boolean	NOT NULL DEFAULT FALSE	Soft-delete

Column	Data Type	Constraints	Description
created_by	uuid	FK auth.users	Creator
updated_by	uuid	FK auth.users	Updater
created_at	timestampz	NOT NULL DEFAULT NOW()	Creation time
updated_at	timestampz	NOT NULL DEFAULT NOW()	Update time

Schema: auth

Authentication, RBAC, JWT token management, sessions.

Table	Description
<code>auth.users</code>	Core identity for all internal staff. Soft-delete via <code>is_deleted</code> . Password is bcrypt-hashed.
<code>auth.roles</code>	Named roles. <code>is_system_role</code> prevents admin deletion of built-in roles.
<code>auth.permissions</code>	Atomic resource:action pairs. System-defined; rarely change without a migration.
<code>auth.user_roles</code>	M:N junction — one user can hold multiple roles.
<code>auth.role_permissions</code>	M:N junction — maps roles to permissions.
<code>auth.groups</code>	Geo/tier support groups. <code>is_lead</code> flag in <code>user_groups</code> grants lead-bucket access.
<code>auth.user_groups</code>	M:N junction. <code>is_lead=true</code> grants agent visibility into lead bucket.
<code>auth.refresh_tokens</code>	SHA-256 hashed refresh tokens with family rotation tracking. On reuse detection, entire <code>family_id</code> is revoked.
<code>auth.token_blacklist</code>	Revoked JWT IDs. Auth middleware performs O(1) lookup on <code>jti</code> before accepting any token.
<code>auth.user_sessions</code>	One row per active login session per device. Supports remote session termination.
<code>auth.user_otp_codes</code>	Hashed OTP codes for email/phone verification and 2FA. Expired rows purged nightly.
<code>auth.user_verification_tokens</code>	Hashed URL tokens for email verification and password reset flows. Single-use.

auth.users

Core identity for all internal staff. Soft-delete via `is_deleted`. Password is bcrypt-hashed.

Column	Data Type	Constraints	Description
<code>id</code>	<code>uuid</code>	PK	Surrogate key
<code>email</code>	<code>varchar(255)</code>	NOT NULL	Raw email address
<code>normalized_email</code>	<code>varchar(255)</code>	NOT NULL UNIQUE	<code>upper(trim(email))</code> for lookups
<code>username</code>	<code>varchar(100)</code>		Optional username
<code>normalized_username</code>	<code>varchar(100)</code>		Normalised username index
<code>password_hash</code>	<code>varchar(255)</code>	NOT NULL	bcrypt hash; never store plain
<code>first_name</code>	<code>varchar(100)</code>		First name
<code>last_name</code>	<code>varchar(100)</code>		Last name
<code>phone</code>	<code>varchar(30)</code>		Phone number
<code>avatar_url</code>	<code>text</code>		Profile photo URL
<code>email_verified</code>	<code>boolean</code>	NOT NULL DEFAULT FALSE	Email verification flag
<code>email_verified_at</code>	<code>timestampz</code>		When email was verified
<code>password_changed_at</code>	<code>timestampz</code>		Last password change
<code>is_active</code>	<code>boolean</code>	NOT NULL DEFAULT TRUE	Active flag

Column	Data Type	Constraints	Description
failed_login_attempts	integer	NOT NULL DEFAULT 0	Brute-force counter
lockout_end	timestampz		Account lockout expiry
last_login_at	timestampz		Last successful login
row_version	bytea		EF Core concurrency token
is_deleted	boolean	NOT NULL DEFAULT FALSE	Soft-delete flag
created_by	uuid	FK auth.users	Creator
updated_by	uuid	FK auth.users	Updater
created_at	timestampz	NOT NULL DEFAULT NOW()	Creation time
updated_at	timestampz	NOT NULL DEFAULT NOW()	Update time

auth.roles

Named roles. is_system_role prevents admin deletion of built-in roles.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
name	varchar(80)	UNIQUE NOT NULL	Role name e.g. junior_agent
description	text		Description
is_system_role	boolean	NOT NULL DEFAULT FALSE	Cannot be deleted via UI
is_deleted	boolean	NOT NULL DEFAULT FALSE	Soft-delete
created_by	uuid	FK auth.users	Creator
updated_by	uuid	FK auth.users	Updater
created_at	timestampz	NOT NULL DEFAULT NOW()	Creation time
updated_at	timestampz		Update time
row_version	bytea		Concurrency token

auth.permissions

Atomic resource:action pairs. System-defined; rarely change without a migration.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
resource	varchar(60)	NOT NULL	Resource name e.g. ticket, company
action	varchar(60)	NOT NULL	Action e.g. create, read, assign
description	text		Human description
is_deleted	boolean	NOT NULL DEFAULT FALSE	Soft-delete
created_at	timestampz	NOT NULL DEFAULT NOW()	Creation time
updated_at	timestampz		Update time
created_by	uuid	FK auth.users	Creator
updated_by	uuid	FK auth.users	Updater
row_version	bytea		Concurrency token

auth.user_roles

M:N junction — one user can hold multiple roles.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
user_id	uuid	NOT NULL FK auth.users CASCADE	User reference
role_id	uuid	NOT NULL FK auth.roles CASCADE	Role reference
assigned_at	timestamptz	NOT NULL DEFAULT NOW()	Assignment time
assigned_by	uuid	FK auth.users	Who assigned the role
is_deleted	boolean	NOT NULL DEFAULT FALSE	Soft-delete
created_at	timestamptz	NOT NULL DEFAULT NOW()	Creation time
updated_at	timestamptz		Update time
created_by	uuid	FK auth.users	Creator
updated_by	uuid	FK auth.users	Updater
row_version	bytea		Concurrency token

auth.role_permissions

M:N junction — maps roles to permissions.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
role_id	uuid	NOT NULL FK auth.roles CASCADE	Role reference
permission_id	uuid	NOT NULL FK auth.permissions CASCADE	Permission reference
is_deleted	boolean	NOT NULL DEFAULT FALSE	Soft-delete
created_at	timestamptz	NOT NULL DEFAULT NOW()	Creation time
updated_at	timestamptz		Update time
created_by	uuid	FK auth.users	Creator
updated_by	uuid	FK auth.users	Updater
row_version	bytea		Concurrency token

auth.groups

Geo/tier support groups. is_lead flag in user_groups grants lead-bucket access.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
name	varchar(100)	NOT NULL UNIQUE	Group name
region_code	varchar(20)	FK lookup.geo_regions	Geographic region
tier_code	varchar(10)	FK lookup.customer_tiers	Customer tier
is_active	boolean	NOT NULL DEFAULT TRUE	Active flag
is_deleted	boolean	NOT NULL DEFAULT FALSE	Soft-delete
created_by	uuid	FK auth.users	Creator
updated_by	uuid	FK auth.users	Updater
created_at	timestamptz	NOT NULL DEFAULT NOW()	Creation time

Column	Data Type	Constraints	Description
updated_at	timestampz		Update time
row_version	bytea		Concurrency token

auth.user_groups

M:N junction. is_lead=true grants agent visibility into lead bucket.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
user_id	uuid	NOT NULL FK auth.users CASCADE	User reference
group_id	uuid	NOT NULL FK auth.groups CASCADE	Group reference
is_lead	boolean	NOT NULL DEFAULT FALSE	Lead flag
is_deleted	boolean	NOT NULL DEFAULT FALSE	Soft-delete
created_by	uuid	FK auth.users	Creator
created_at	timestampz	NOT NULL DEFAULT NOW()	Creation time
row_version	bytea		Concurrency token

auth.refresh_tokens

SHA-256 hashed refresh tokens with family rotation tracking. On reuse detection, entire family_id is revoked.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
user_id	uuid	NOT NULL FK auth.users CASCADE	Owner user
token_hash	varchar(255)	UNIQUE NOT NULL	SHA-256 of raw token
family_id	uuid	NOT NULL	Rotation family ID
device_info	varchar(255)		Device description
ip_address	inet		Client IP
issued_at	timestampz	NOT NULL DEFAULT NOW()	Issue time
expires_at	timestampz	NOT NULL	Expiry time
last_used_at	timestampz		Last use time
rotated_at	timestampz		Rotation time
is_revoked	boolean	NOT NULL DEFAULT FALSE	Revocation flag
revoked_at	timestampz		Revocation time
revoked_reason	auth.revocation_reason		Revocation reason enum
replaced_by_token_id	uuid	FK auth.refresh_tokens	Successor token
created_by_ip	varchar(45)		Creation client IP
revoked_by_ip	varchar(45)		Revocation client IP
user_agent	text		User-agent string
created_at	timestampz	NOT NULL DEFAULT NOW()	Creation time

auth.token_blacklist

Revoked JWT IDs. Auth middleware performs O(1) lookup on jti before accepting any token.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
jti	varchar(36)	UNIQUE NOT NULL	JWT ID claim
user_id	uuid	NOT NULL FK auth.users CASCADE	Token owner
expires_at	timestampz	NOT NULL	Token expiry for pruning
reason	varchar(100)		Revocation reason
blacklisted_at	timestampz	NOT NULL DEFAULT NOW()	Blacklist time

auth.user_sessions

One row per active login session per device. Supports remote session termination.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
user_id	uuid	NOT NULL FK auth.users CASCADE	Owner user
refresh_token_id	uuid	FK auth.refresh_tokens	Linked token
device_name	varchar(150)		Device description
ip_address	inet		Client IP
geo_location	varchar(100)		Approximate location
started_at	timestampz	NOT NULL DEFAULT NOW()	Session start
last_active_at	timestampz	NOT NULL DEFAULT NOW()	Last API call
ended_at	timestampz		Session end
is_active	boolean	NOT NULL DEFAULT TRUE	Active flag

auth.user_otp_codes

Hashed OTP codes for email/phone verification and 2FA. Expired rows purged nightly.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
user_id	uuid	NOT NULL FK auth.users CASCADE	Owner user
code_hash	varchar(255)	NOT NULL	bcrypt/SHA-256 of raw OTP
purpose	varchar(60)	NOT NULL	email_verify phone_verify two_factor password_reset
delivery_target	varchar(255)		Email or phone OTP was sent to
expires_at	timestampz	NOT NULL	OTP expiry
created_at	timestampz	NOT NULL DEFAULT NOW()	Creation time
verified_at	timestampz		When OTP was verified
failed_attempts	integer	NOT NULL DEFAULT 0	Wrong-guess counter
is_used	boolean	NOT NULL DEFAULT FALSE	Used flag

auth.user_verification_tokens

Hashed URL tokens for email verification and password reset flows. Single-use.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key

Column	Data Type	Constraints	Description
user_id	uuid	NOT NULL FK auth.users CASCADE	Owner user
token_hash	varchar(255)	UNIQUE NOT NULL	SHA-256 of raw URL token
purpose	varchar(60)	NOT NULL	email_verify password_reset invite magic_link
expires_at	timestampz	NOT NULL	Token expiry
created_at	timestampz	NOT NULL DEFAULT NOW()	Creation time
verified_at	timestampz		When token was used
is_used	boolean	NOT NULL DEFAULT FALSE	Single-use flag
created_by_ip	varchar(45)		Requesting client IP

Schema: company

Customer accounts, domains, contacts, limits.

Table	Description
<code>company.companies</code>	Core customer account. <code>health_score</code> + <code>priority_bump</code> elevates ticket priority. <code>delivery_support_active</code> routes to delivery (hypercare). <code>payment_on_hold</code> triggers a
<code>company.company_domains</code>	Email domains per company. UNIQUE on lower(domain) enables O(1) auto-routing of incoming emails.
<code>company.contacts</code>	Authorized contacts per company. <code>is_authorized</code> gates ticket creation. <code>auto_created</code> for smart inbound email routing.
<code>company.company_contacts_limit</code>	Configurable per-company override for max authorized contacts. One config per company.

company.companies

Core customer account. `health_score` + `priority_bump` elevates ticket priority. `delivery_support_active` routes to delivery (hypercare). `payment_on_hold` triggers auto On Hold.

Column	Data Type	Constraints	Description
<code>id</code>	<code>uuid</code>	PK	Surrogate key
<code>name</code>	<code>varchar(200)</code>	NOT NULL	Company name
<code>csp_id</code>	<code>varchar(100)</code>	UNIQUE	CSP identifier
<code>deployment_model</code>	<code>company.deployment_model</code>	NOT NULL DEFAULT cloud	cloud on_premise hybrid
<code>geo_region</code>	<code>varchar(20)</code>	NOT NULL FK lookup.geo_regions	Region code
<code>support_tier</code>	<code>varchar(10)</code>	NOT NULL FK lookup.customer_tier	Tier code
<code>industry</code>	<code>varchar(100)</code>		Industry vertical
<code>health_score</code>	<code>integer</code>	NOT NULL DEFAULT 75 (0-100)	Customer health score
<code>is_active</code>	<code>boolean</code>	NOT NULL DEFAULT TRUE	Active flag
<code>is_payroll_customer</code>	<code>boolean</code>	NOT NULL DEFAULT FALSE	Payroll module flag
<code>leave_credit_cycle</code>	<code>varchar(30)</code>		Leave accrual cycle
<code>migration_date</code>	<code>date</code>		Migration date
<code>cam_user_id</code>	<code>uuid</code>	FK auth.users	Customer Account Manager
<code>delivery_manager_id</code>	<code>uuid</code>	FK auth.users	Delivery Manager
<code>delivery_support_active</code>	<code>boolean</code>	NOT NULL DEFAULT FALSE	Hypercare mode flag
<code>payment_on_hold</code>	<code>boolean</code>	NOT NULL DEFAULT FALSE	Financial hold flag
<code>notes</code>	<code>text</code>		Internal notes
<code>is_deleted</code>	<code>boolean</code>	NOT NULL DEFAULT FALSE	Soft-delete
<code>created_by</code>	<code>uuid</code>	FK auth.users	Creator
<code>updated_by</code>	<code>uuid</code>	FK auth.users	Updater
<code>created_at</code>	<code>timestampz</code>	NOT NULL DEFAULT NOW()	Creation time
<code>updated_at</code>	<code>timestampz</code>	NOT NULL DEFAULT NOW()	Update time

company.company_domains

Email domains per company. UNIQUE on lower(domain) enables O(1) auto-routing of incoming emails.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
company_id	uuid	NOT NULL FK company.companies ON SCALAR CASCADE	Parent company
domain	varchar(255)	NOT NULL UNIQUE	Email domain e.g. acme.com
is_primary	boolean	NOT NULL DEFAULT FALSE	Primary domain flag
is_deleted	boolean	NOT NULL DEFAULT FALSE	Soft-delete
created_at	timestampz	NOT NULL DEFAULT NOW()	Creation time

company.contacts

Authorized contacts per company. is_authorized gates ticket creation. auto_created for smart inbound email routing.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
company_id	uuid	NOT NULL FK company.companies ON SCALAR CASCADE	Parent company
user_id	uuid	FK auth.users	Linked staff user (if any)
name	varchar(200)	NOT NULL	Contact full name
email	varchar(255)	NOT NULL UNIQUE	Contact email
phone	varchar(30)		Contact phone
is_authorized	boolean	NOT NULL DEFAULT TRUE	Can raise tickets
auto_created	boolean	NOT NULL DEFAULT FALSE	System auto-created flag
is_deleted	boolean	NOT NULL DEFAULT FALSE	Soft-delete
created_by	uuid	FK auth.users	Creator
updated_by	uuid	FK auth.users	Updater
created_at	timestampz	NOT NULL DEFAULT NOW()	Creation time
updated_at	timestampz	NOT NULL DEFAULT NOW()	Update time

company.company_contacts_limit

Configurable per-company override for max authorized contacts. One config per company.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
company_id	uuid	UNIQUE NOT NULL FK company.companies ON SCALAR CASCADE	Parent company
max_contacts	integer	NOT NULL DEFAULT 10 (1-100)	Max authorized contacts
created_by	uuid	FK auth.users	Creator
created_at	timestampz	NOT NULL DEFAULT NOW()	Creation time

Schema: workflow

Graph-based ticket status transition engine.

Table	Description
<code>workflow.ticket_status_graphs</code>	Named status transition graphs. <code>graph_id</code> resolved once at ticket creation and stored on ticket.
<code>workflow.status_transitions</code>	One row per allowed edge in a graph (<code>from_status</code> to <code>to_status</code>). NULL <code>from_status</code> = entry point.
<code>workflow.ticket_status_graph_scope_s</code>	Scope-to-graph mapping. NULL dimensions = wildcards. Resolver picks highest priority match.

workflow.ticket_status_graphs

Named status transition graphs. `graph_id` resolved once at ticket creation and stored on ticket.

Column	Data Type	Constraints	Description
<code>id</code>	<code>uuid</code>	PK	Surrogate key
<code>graph_code</code>	<code>varchar(40)</code>	UNIQUE NOT NULL	Stable key e.g. GRAPH-001
<code>name</code>	<code>varchar(150)</code>	NOT NULL	Human-readable name
<code>description</code>	<code>text</code>		Description
<code>is_active</code>	<code>boolean</code>	NOT NULL DEFAULT TRUE	Active flag
<code>is_deleted</code>	<code>boolean</code>	NOT NULL DEFAULT FALSE	Soft-delete
<code>created_by</code>	<code>uuid</code>	FK auth.users	Creator
<code>updated_by</code>	<code>uuid</code>	FK auth.users	Updater
<code>created_at</code>	<code>timestampz</code>	NOT NULL DEFAULT NOW()	Creation time
<code>updated_at</code>	<code>timestampz</code>	NOT NULL DEFAULT NOW()	Update time

workflow.status_transitions

One row per allowed edge in a graph (`from_status` to `to_status`). NULL `from_status` = entry point.

Column	Data Type	Constraints	Description
<code>id</code>	<code>uuid</code>	PK	Surrogate key
<code>graph_id</code>	<code>uuid</code>	NOT NULL FK workflow.ticket_status_graphs	Parent graph
<code>from_status</code>	<code>varchar(40)</code>		Source status; NULL = initial entry
<code>to_status</code>	<code>varchar(40)</code>	NOT NULL	Target status
<code>requires_role</code>	<code>varchar(80)</code>	FK auth.roles	Only this role may trigger
<code>requires_field</code>	<code>varchar(100)</code>		Named ticket column must be non-null
<code>auto_trigger</code>	<code>boolean</code>	NOT NULL DEFAULT FALSE	Engine can fire without human actor
<code>display_order</code>	<code>integer</code>	NOT NULL DEFAULT 0	UI ordering
<code>is_deleted</code>	<code>boolean</code>	NOT NULL DEFAULT FALSE	Soft-delete
<code>created_by</code>	<code>uuid</code>	FK auth.users	Creator
<code>created_at</code>	<code>timestampz</code>	NOT NULL DEFAULT NOW()	Creation time

workflow.ticket_status_graph_scopes

Scope-to-graph mapping. NULL dimensions = wildcards. Resolver picks highest priority match.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
graph_id	uuid	NOT NULL FK workflow.ticket_status_graphs	Target graph
department	varchar(80)		NULL = any department
version_id	uuid	FK lookup.product_versions	NULL = any version
module_id	uuid	FK lookup.modules	NULL = any module
sub_module_id	uuid	FK lookup.sub_modules	NULL = any sub-module
priority	integer	NOT NULL DEFAULT 0	Higher = evaluated first
is_active	boolean	NOT NULL DEFAULT TRUE	Active flag
is_deleted	boolean	NOT NULL DEFAULT FALSE	Soft-delete
created_by	uuid	FK auth.users	Creator
updated_by	uuid	FK auth.users	Updater
created_at	timestampz	NOT NULL DEFAULT NOW()	Creation time
updated_at	timestampz	NOT NULL DEFAULT NOW()	Update time

Schema: ticket

Central transactional domain — tickets, comments, attachments, history, assignments, CSAT, custom fields.

Table	Description
<code>ticket.tickets</code>	Central transactional entity. <code>graph_id</code> resolved once at creation. <code>version_id/module_id/sub_module_id</code> are normalized FK refs.
<code>ticket.ticket_comments</code>	Threaded conversation. <code>is_private=true</code> = internal note hidden from customer portal.
<code>ticket.ticket_attachments</code>	Object storage references. <code>file_url</code> points to S3/Azure Blob. <code>comment_id</code> nullable.
<code>ticket.ticket_status_history</code>	IMMUTABLE append-only audit log of every status transition. Never update or delete rows.
<code>ticket.ticket_assignment_log</code>	Every reassignment recorded. <code>changed_by</code> = lead/manager who performed the action.
<code>ticket.csat_surveys</code>	One CSAT per ticket. Triggered on customer close. Rating 1-5.
<code>ticket.ticket_custom_fields</code>	EAV extension. Allows custom fields without schema migrations. Useful for graph-specific metadata.

ticket.tickets

Central transactional entity. `graph_id` resolved once at creation. `version_id/module_id/sub_module_id` are normalized FK refs.

Column	Data Type	Constraints	Description
<code>id</code>	<code>uuid</code>	PK	Surrogate key
<code>ticket_number</code>	<code>varchar(20)</code>	UNIQUE	Auto-generated TKT-YYYYMM-NNNNN
<code>company_id</code>	<code>uuid</code>	NOT NULL FK <code>company.companies</code>	Owning company
<code>contact_id</code>	<code>uuid</code>	FK <code>company.contacts</code>	Raising contact
<code>group_id</code>	<code>uuid</code>	FK <code>auth.groups</code>	Assigned group
<code>assigned_agent_id</code>	<code>uuid</code>	FK <code>auth.users</code>	Assigned agent
<code>created_by_user_id</code>	<code>uuid</code>	FK <code>auth.users</code>	Staff who created ticket
<code>graph_id</code>	<code>uuid</code>	FK <code>workflow.ticket_status_graphs</code>	Status graph in use
<code>version_id</code>	<code>uuid</code>	FK <code>lookup.product_versions</code>	Product version
<code>module_id</code>	<code>uuid</code>	NOT NULL FK <code>lookup.modules</code>	Module
<code>sub_module_id</code>	<code>uuid</code>	FK <code>lookup.sub_modules</code>	Sub-module
<code>source</code>	<code>ticket.ticket_source</code>	NOT NULL DEFAULT <code>portal</code>	<code>email portal phone</code>
<code>priority</code>	<code>ticket.ticket_priority</code>	NOT NULL DEFAULT <code>medium</code>	<code>urgent high medium low</code>
<code>status</code>	<code>ticket.ticket_status</code>	NOT NULL DEFAULT <code>new</code>	Current status
<code>type</code>	<code>ticket.ticket_type</code>	NOT NULL DEFAULT <code>query</code>	<code>bug enhancement incident ...</code>
<code>subject</code>	<code>text</code>	NOT NULL	Ticket subject
<code>description</code>	<code>text</code>	NOT NULL	Ticket body
<code>product_type</code>	<code>varchar(40)</code>		<code>Customization Geo-specific Base</code>
<code>solution_type</code>	<code>varchar(40)</code>		<code>Data correction Patch Config ...</code>

Column	Data Type	Constraints	Description
fix_type	varchar(20)		Permanent Temporary
rca	text		Root cause analysis (mandatory for closure)
customer_call_taken	boolean	NOT NULL DEFAULT FALSE	Customer call flag
linked_jira_id	varchar(100)		Linked Jira issue
ai_rating	ai.agent_rating		AI suggestion rating
audited_by	uuid	FK auth.users	Auditing manager
audit_notes	text		Audit notes
sla_excluded	boolean	NOT NULL DEFAULT FALSE	SLA exclusion flag
is_on_hold_payment	boolean	NOT NULL DEFAULT FALSE	Payment hold flag
is_deleted	boolean	NOT NULL DEFAULT FALSE	Soft-delete
created_by	uuid	FK auth.users	Creator
updated_by	uuid	FK auth.users	Updater
created_at	timestampz	NOT NULL DEFAULT NOW()	Creation time
updated_at	timestampz	NOT NULL DEFAULT NOW()	Update time

ticket.ticket_comments

Threaded conversation. is_private=true = internal note hidden from customer portal.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
ticket_id	uuid	NOT NULL FK ticket.tickets CASCADE	Parent ticket
author_id	uuid	FK auth.users	Staff author
contact_id	uuid	FK company.contacts	Customer author
body	text	NOT NULL	Comment body
is_private	boolean	NOT NULL DEFAULT FALSE	Internal note flag
is_deleted	boolean	NOT NULL DEFAULT FALSE	Soft-delete
created_by	uuid	FK auth.users	Creator
created_at	timestampz	NOT NULL DEFAULT NOW()	Creation time
updated_at	timestampz	NOT NULL DEFAULT NOW()	Update time

ticket.ticket_attachments

Object storage references. file_url points to S3/Azure Blob. comment_id nullable.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
ticket_id	uuid	NOT NULL FK ticket.tickets CASCADE	Parent ticket
comment_id	uuid	FK ticket.ticket_comments	Linked comment (optional)
file_name	varchar(255)	NOT NULL	Original file name
file_url	text	NOT NULL	Object storage URL
file_size_bytes	bigint		File size in bytes
mime_type	varchar(100)		MIME type

Column	Data Type	Constraints	Description
is_deleted	boolean	NOT NULL DEFAULT FALSE	Soft-delete
created_by	uuid	FK auth.users	Uploader
created_at	timestampz	NOT NULL DEFAULT NOW()	Upload time

ticket.ticket_status_history

IMMUTABLE append-only audit log of every status transition. Never update or delete rows.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
ticket_id	uuid	NOT NULL FK ticket.tickets CASCADE	Parent ticket
from_status	varchar(40)		Previous status (NULL = initial)
to_status	varchar(40)	NOT NULL	New status
changed_by	uuid	FK auth.users	Actor who changed status
changed_at	timestampz	NOT NULL DEFAULT NOW()	Transition time
reason	text		Optional reason

ticket.ticket_assignment_log

Every reassignment recorded. changed_by = lead/manager who performed the action.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
ticket_id	uuid	NOT NULL FK ticket.tickets CASCADE	Parent ticket
from_agent_id	uuid	FK auth.users	Previous agent
to_agent_id	uuid	NOT NULL FK auth.users	New agent
changed_by	uuid	FK auth.users	Actor
assigned_at	timestampz	NOT NULL DEFAULT NOW()	Assignment time
notes	text		Reason notes

ticket.csat_surveys

One CSAT per ticket. Triggered on customer close. Rating 1-5.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
ticket_id	uuid	UNIQUE NOT NULL FK ticket.tickets CASCADE	Parent ticket
contact_id	uuid	FK company.contacts	Responding contact
rating	integer	NOT NULL CHECK (1-5)	Star rating
feedback	text		Free-text feedback
submitted_at	timestampz	NOT NULL DEFAULT NOW()	Submission time

ticket.ticket_custom_fields

EAV extension. Allows custom fields without schema migrations. Useful for graph-specific metadata.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
ticket_id	uuid	NOT NULL FK ticket.tickets CASCADE	Parent ticket
field_key	varchar(80)	NOT NULL	Custom field key
field_value	text		Custom field value
created_at	timestampz	NOT NULL DEFAULT NOW()	Creation time

Schema: sla

SLA policies, per-ticket clocks, business hours, holidays, automation rules, escalation ladders.

Table	Description
<code>sla.sla_policies</code>	One policy per (geo_region x tier_code x priority). SLA minutes are business-hours only.
<code>sla.sla_tickets</code>	Per-ticket SLA clock. One-to-one with tickets. paused_minutes accumulates total pause time.
<code>sla.business_hours</code>	Business hours per geo region. SLA clock only ticks within these hours.
<code>sla.holidays</code>	Region-specific public holidays. SLA engine skips these dates.
<code>sla.automation_rules</code>	JSONB conditions and actions. Triggers: ticket_created, ticket_updated, time_based. GIN indexed.
<code>sla.automation_execution_log</code>	Immutable audit of every rule execution per ticket. Append-only; never update or delete.
<code>sla.escalation_rules</code>	Time-triggered no-response ladders: 2h→agent, 4h→lead, 24h→agent, 48h→manager.

sla.sla_policies

One policy per (geo_region x tier_code x priority). SLA minutes are business-hours only.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
name	varchar(100)	NOT NULL	Policy name
geo_region	varchar(20)	NOT NULL FK lookup.geo_regions	Region code
tier_code	varchar(10)	NOT NULL FK lookup.customer_tier	Tier code
priority	varchar(20)	NOT NULL CHECK urgent high medium low	Priority level
first_response_minutes	integer	NOT NULL	First response SLA (biz mins)
resolution_minutes	integer	NOT NULL	Resolution SLA (biz mins)
is_active	boolean	NOT NULL DEFAULT TRUE	Active flag
is_deleted	boolean	NOT NULL DEFAULT FALSE	Soft-delete
created_by	uuid	FK auth.users	Creator
updated_by	uuid	FK auth.users	Updater
created_at	timestampz	NOT NULL DEFAULT NOW()	Creation time
updated_at	timestampz	NOT NULL DEFAULT NOW()	Update time

sla.sla_tickets

Per-ticket SLA clock. One-to-one with tickets. paused_minutes accumulates total pause time.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
ticket_id	uuid	UNIQUE NOT NULL FK ticket.tickets	Parent ticket (1:1)
policy_id	uuid	NOT NULL FK sla.sla_policies	Applied policy
first_response_due_at	timestampz	NOT NULL	First response deadline

Column	Data Type	Constraints	Description
first_response_at	timestampz		Actual first response time
resolution_due_at	timestampz	NOT NULL	Resolution deadline
resolved_at	timestampz		Actual resolution time
first_response_breached	boolean	NOT NULL DEFAULT FALSE	First response breach flag
resolution_breached	boolean	NOT NULL DEFAULT FALSE	Resolution breach flag
paused_minutes	integer	NOT NULL DEFAULT 0	Total pause duration
last_paused_at	timestampz		Last pause start
created_at	timestampz	NOT NULL DEFAULT NOW()	Creation time
updated_at	timestampz	NOT NULL DEFAULT NOW()	Update time

sla.business_hours

Business hours per geo region. SLA clock only ticks within these hours.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
geo_region	varchar(20)	NOT NULL FK lookup.geo_regions	Region code
start_time	time	NOT NULL	Business start time
end_time	time	NOT NULL	Business end time
working_days	varchar(20)	NOT NULL DEFAULT Mon-Fri	Working day range
is_deleted	boolean	NOT NULL DEFAULT FALSE	Soft-delete
created_by	uuid	FK auth.users	Creator
created_at	timestampz	NOT NULL DEFAULT NOW()	Creation time

sla.holidays

Region-specific public holidays. SLA engine skips these dates.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
geo_region	varchar(20)	NOT NULL FK lookup.geo_regions	Region code
holiday_date	date	NOT NULL	Holiday date
name	varchar(100)	NOT NULL	Holiday name
is_deleted	boolean	NOT NULL DEFAULT FALSE	Soft-delete
created_by	uuid	FK auth.users	Creator
created_at	timestampz	NOT NULL DEFAULT NOW()	Creation time

sla.automation_rules

JSONB conditions and actions. Triggers: ticket_created, ticket_updated, time_based. GIN indexed.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
name	varchar(150)	NOT NULL	Rule name
trigger_event	sla.trigger_event	NOT NULL	ticket_created ticket_updated time_based

Column	Data Type	Constraints	Description
conditions	jsonb	NOT NULL DEFAULT []	Rule conditions array
actions	jsonb	NOT NULL DEFAULT []	Rule actions array
execution_order	integer	NOT NULL DEFAULT 100	Evaluation sequence
is_active	boolean	NOT NULL DEFAULT TRUE	Active flag
is_deleted	boolean	NOT NULL DEFAULT FALSE	Soft-delete
created_by	uuid	FK auth.users	Creator
updated_by	uuid	FK auth.users	Updater
created_at	timestamptz	NOT NULL DEFAULT NOW()	Creation time
updated_at	timestamptz	NOT NULL DEFAULT NOW()	Update time

sla.automation_execution_log

Immutable audit of every rule execution per ticket. Append-only; never update or delete.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
rule_id	uuid	NOT NULL FK sla.automation_rules	Foreign key to rule_id
ticket_id	uuid	NOT NULL FK ticket.tickets CASCADE	Foreign key to ticket_id
succeeded	boolean	NOT NULL DEFAULT TRUE	Execution success flag
error_message	text		Error detail on failure
executed_at	timestamptz	NOT NULL DEFAULT NOW()	Execution time

sla.escalation_rules

Time-triggered no-response ladders: 2h→agent, 4h→lead, 24h→agent, 48h→manager.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
name	varchar(100)	NOT NULL	Rule name
no_response_minutes	integer	NOT NULL	Threshold in minutes
notify_role	varchar(60)	NOT NULL	Role to notify
is_active	boolean	NOT NULL DEFAULT TRUE	Active flag
created_by	uuid	FK auth.users	Creator
created_at	timestamptz	NOT NULL DEFAULT NOW()	Creation time

Schema: kb

Knowledge base folders, articles, attachments, portal banners.

Table	Description
kb.kb_folders	Self-referencing folder hierarchy. parent_id=NULL for root folders. Use WITH RECURSIVE CTE for full tree.
kb.kb_articles	Release notes, FAQs, user manuals, patches, process flows.
kb.kb_attachments	Files attached to KB articles. Stored in object storage; URL reference only.
kb.portal_banners	Time-boxed banners on the customer portal. active_from/active_to = schedule. is_active = manual override.

kb.kb_folders

Self-referencing folder hierarchy. parent_id=NULL for root folders. Use WITH RECURSIVE CTE for full tree.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
name	varchar(150)	NOT NULL	Folder name
parent_id	uuid	FK kb.kb_folders	Parent folder (NULL = root)
display_order	integer	NOT NULL DEFAULT 0	UI ordering
is_deleted	boolean	NOT NULL DEFAULT FALSE	Soft-delete
created_by	uuid	FK auth.users	Creator
updated_by	uuid	FK auth.users	Updater
created_at	timestamptz	NOT NULL DEFAULT NOW()	Creation time
updated_at	timestamptz	NOT NULL DEFAULT NOW()	Update time

kb.kb_articles

Release notes, FAQs, user manuals, patches, process flows.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
title	varchar(300)	NOT NULL	Article title
content	text		Article body (HTML/Markdown)
article_type	varchar(40)	NOT NULL CHECK faq release_note user_manual patch process_flow	Article category
status	varchar(30)	NOT NULL DEFAULT draft CHECK draft published archived	Article status
author_id	uuid	FK auth.users	Article author
folder_id	uuid	FK kb.kb_folders	Parent folder
is_published	boolean	NOT NULL DEFAULT FALSE	Published flag
is_deleted	boolean	NOT NULL DEFAULT FALSE	Soft-delete
created_by	uuid	FK auth.users	Creator
updated_by	uuid	FK auth.users	Updater
created_at	timestamptz	NOT NULL DEFAULT NOW()	Creation time
updated_at	timestamptz	NOT NULL DEFAULT NOW()	Update time

kb.kb_attachments

Files attached to KB articles. Stored in object storage; URL reference only.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
article_id	uuid	NOT NULL FK kb.kb_articles CASCADE	Parent article
file_url	text	NOT NULL	Object storage URL
file_name	varchar(255)	NOT NULL	Original file name
file_size_bytes	bigint		File size in bytes
mime_type	varchar(100)		MIME type
is_deleted	boolean	NOT NULL DEFAULT FALSE	Soft-delete
created_at	timestampz	NOT NULL DEFAULT NOW()	Creation time

kb.portal_banners

Time-boxed banners on the customer portal. active_from/active_to = schedule. is_active = manual override.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
title	varchar(200)	NOT NULL	Banner title
message	text	NOT NULL	Banner body
active_from	timestampz		Schedule start
active_to	timestampz		Schedule end
is_active	boolean	NOT NULL DEFAULT TRUE	Manual override flag
created_by	uuid	FK auth.users	Creator
updated_by	uuid	FK auth.users	Updater
created_at	timestampz	NOT NULL DEFAULT NOW()	Creation time
updated_at	timestampz	NOT NULL DEFAULT NOW()	Update time

Schema: notification

Handlebars templates, delivery logs, failed delivery tracking.

Table	Description
<code>notification.notification_templates</code>	Handlebars templates for all notification events. code is the stable reference key used by automation.
<code>notification.notification_logs</code>	Delivery audit. is_failed_delivery=true surfaces on dashboard widget. ticket_id nullable for system notifications.

notification.notification_templates

Handlebars templates for all notification events. code is the stable reference key used by automation.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
code	varchar(80)	UNIQUE NOT NULL	Event code e.g. TICKET_CREATED
name	varchar(150)	NOT NULL	Template name
channel	notification.notification_channels	NOT NULL DEFAULT email	email in_app sms
subject	varchar(300)		Email subject (Handlebars)
body_html	text		HTML body (Handlebars)
is_active	boolean	NOT NULL DEFAULT TRUE	Active flag
is_deleted	boolean	NOT NULL DEFAULT FALSE	Soft-delete
created_by	uuid	FK auth.users	Creator
updated_by	uuid	FK auth.users	Updater
created_at	timestampz	NOT NULL DEFAULT NOW()	Creation time
updated_at	timestampz	NOT NULL DEFAULT NOW()	Update time

notification.notification_logs

Delivery audit. is_failed_delivery=true surfaces on dashboard widget. ticket_id nullable for system notifications.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
ticket_id	uuid	FK ticket.tickets	Related ticket (nullable)
template_id	uuid	NOT NULL FK notification.notification_templates	Used template
recipient_email	varchar(255)	NOT NULL	Recipient email address
status	notification.delivery_statuses	NOT NULL DEFAULT pending	sent failed bounced pending
error_message	text		Delivery error detail
is_failed_delivery	boolean	NOT NULL DEFAULT FALSE	Failed delivery flag
sent_at	timestampz	NOT NULL DEFAULT NOW()	Send time

Schema: ai

AI suggestion logs with feedback loop ratings.

Table	Description
ai.ai_suggestion_logs	AI suggestions shown to agent on new ticket. Two rows per ticket: troubleshooting_steps and similar_tickets.

ai.ai_suggestion_logs

AI suggestions shown to agent on new ticket. Two rows per ticket: troubleshooting_steps and similar_tickets.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
ticket_id	uuid	NOT NULL FK ticket.tickets CASCADE	Parent ticket
agent_id	uuid	FK auth.users	Viewing agent
suggestion_type	ai.suggestion_type	NOT NULL	troubleshooting_steps similar_tickets
suggestion_content	text	NOT NULL	AI-generated suggestion text
agent_rating	ai.agent_rating		relevant moderately_relevant not_relevant
created_at	timestampz	NOT NULL DEFAULT NOW()	Creation time

Schema: gamification

Points, badges, streaks, leaderboards, challenges.

Table	Description
<code>gamification.badges</code>	Badge definitions. tier = bronze/silver/gold/platinum. points_value = bonus on award.
<code>gamification.point_rules</code>	Point-earning rule definitions. max_per_day prevents farming.
<code>gamification.agent_points</code>	IMMUTABLE append-only point ledger. Event-sourcing pattern. Total = SUM(points). Never update or delete.
<code>gamification.agent_badges</code>	Badge award records. notified=false triggers push notification.
<code>gamification.leaderboard_snapshots</code>	Pre-computed daily/weekly/monthly leaderboards. Rebuilt at 00:05 by scheduled job.
<code>gamification.agent_streaks</code>	One row per agent. Consecutive-day resolution streaks. Nightly job resets if last_activity_date < TODAY-1.
<code>gamification.challenges</code>	Time-boxed gamification events. scope=global auto-enrolls all agents.
<code>gamification.agent_challenges</code>	Per-agent progress on active challenges. current_value incremented by gamification engine.

gamification.badges

Badge definitions. tier = bronze/silver/gold/platinum. points_value = bonus on award.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
code	varchar(80)	UNIQUE NOT NULL	Badge code
name	varchar(150)	NOT NULL	Badge name
description	text		Description
icon_url	text		Badge icon URL
tier	gamification.badge_tier	NOT NULL DEFAULT bronze	bronze silver gold platinum
points_value	integer	NOT NULL DEFAULT 0	Bonus points on award
is_active	boolean	NOT NULL DEFAULT TRUE	Active flag
is_deleted	boolean	NOT NULL DEFAULT FALSE	Soft-delete
created_by	uuid	FK auth.users	Creator
created_at	timestampz	NOT NULL DEFAULT NOW()	Creation time

gamification.point_rules

Point-earning rule definitions. max_per_day prevents farming.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
code	varchar(80)	UNIQUE NOT NULL	Rule code e.g. FIRST_RESPONSE_ON_TIME
description	text		Description

Column	Data Type	Constraints	Description
points	integer	NOT NULL	Points awarded (negative for reversals)
max_per_day	integer		Daily cap; NULL = unlimited
is_active	boolean	NOT NULL DEFAULT TRUE	Active flag
created_by	uuid	FK auth.users	Creator
created_at	timestamptz	NOT NULL DEFAULT NOW()	Creation time
updated_at	timestamptz	NOT NULL DEFAULT NOW()	Update time

gamification.agent_points

IMMUTABLE append-only point ledger. Event-sourcing pattern. Total = SUM(points). Never update or delete.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
agent_id	uuid	NOT NULL FK auth.users CASCADE	Point earner
rule_id	uuid	NOT NULL FK gamification.point_rules	Triggering rule
ticket_id	uuid	FK ticket.tickets	Related ticket (nullable)
points	integer	NOT NULL	Points delta
description	varchar(200)		Human description
earned_at	timestamptz	NOT NULL DEFAULT NOW()	Earn time
earned_date	date	NOT NULL DEFAULT today	UTC date for daily cap check

gamification.agent_badges

Badge award records. notified=false triggers push notification.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
agent_id	uuid	NOT NULL FK auth.users CASCADE	Award recipient
badge_id	uuid	NOT NULL FK gamification.badges CASCADE	Awarded badge
ticket_id	uuid	FK ticket.tickets	Related ticket (nullable)
earned_at	timestamptz	NOT NULL DEFAULT NOW()	Award time
notified	boolean	NOT NULL DEFAULT FALSE	Notification sent flag

gamification.leaderboard_snapshots

Pre-computed daily/weekly/monthly leaderboards. Rebuilt at 00:05 by scheduled job.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
agent_id	uuid	NOT NULL FK auth.users CASCADE	Agent
group_id	uuid	FK auth.groups	Group scope (nullable)
period_type	gamification.period_type	NOT NULL	daily weekly monthly all_time
period_start	date	NOT NULL	Period start date
period_end	date	NOT NULL	Period end date
total_points	integer	NOT NULL DEFAULT 0	Total points in period

Column	Data Type	Constraints	Description
tickets_resolved	integer	NOT NULL DEFAULT 0	Tickets resolved count
sla_met_count	integer	NOT NULL DEFAULT 0	SLA met count
avg_csat	numeric(4,2)		Average CSAT rating
rank	integer		Rank in period
computed_at	timestampz	NOT NULL DEFAULT NOW()	Computation time

gamification.agent_streaks

One row per agent. Consecutive-day resolution streaks. Nightly job resets if last_activity_date < TODAY-1.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
agent_id	uuid	UNIQUE NOT NULL FK auth.users ON DELETE CASCADE	Agent ID
current_streak_days	integer	NOT NULL DEFAULT 0	Current streak length
longest_streak_days	integer	NOT NULL DEFAULT 0	All-time longest streak
streak_start_date	date		Current streak start date
last_activity_date	date		Last qualifying activity date
updated_at	timestampz	NOT NULL DEFAULT NOW()	Update time

gamification.challenges

Time-boxed gamification events. scope=global auto-enrolls all agents.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
title	varchar(200)	NOT NULL	Challenge title
description	text		Description
metric	varchar(80)	NOT NULL	Metric being measured
target_value	integer	NOT NULL	Target count to complete
bonus_points	integer	NOT NULL DEFAULT 0	Points on completion
badge_id	uuid	FK gamification.badges	Badge awarded on completion
scope	varchar(20)	NOT NULL DEFAULT global CHECK (scope IN ('global', 'individual'))	Challenge scope
group_id	uuid	FK auth.groups	Target group (for group scope)
starts_at	timestampz	NOT NULL	Challenge start
ends_at	timestampz	NOT NULL	Challenge end
is_active	boolean	NOT NULL DEFAULT TRUE	Active flag
created_by	uuid	FK auth.users	Creator
created_at	timestampz	NOT NULL DEFAULT NOW()	Creation time

gamification.agent_challenges

Per-agent progress on active challenges. current_value incremented by gamification engine.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key

Column	Data Type	Constraints	Description
agent_id	uuid	NOT NULL FK auth.users CASCADE	Agent
challenge_id	uuid	NOT NULL FK gamification.challenges CASCADE	Challenge
current_value	integer	NOT NULL DEFAULT 0	Progress count
is_completed	boolean	NOT NULL DEFAULT FALSE	Completion flag
completed_at	timestampz		Completion time
points_awarded	integer	NOT NULL DEFAULT 0	Points actually awarded

Schema: audit

Cross-domain audit log for compliance and forensic investigation.

Table	Description
<code>audit.audit_log</code>	Cross-schema forensic audit log. <code>old_values/new_values</code> store JSONB diff for immutable change history.

audit.audit_log

Cross-schema forensic audit log. `old_values/new_values` store JSONB diff for immutable change history.

Column	Data Type	Constraints	Description
id	uuid	PK	Surrogate key
schema_name	varchar(60)	NOT NULL	Source schema
table_name	varchar(60)	NOT NULL	Source table
record_id	uuid	NOT NULL	Modified record ID
action	varchar(10)	NOT NULL CHECK INSERT UPDATE DELETE	Operation type
old_values	jsonb		Previous row values
new_values	jsonb		New row values
changed_by	uuid	FK auth.users	Actor
changed_at	timestampz	NOT NULL DEFAULT NOW()	Change time
ip_address	inet		Client IP
session_id	uuid		Session reference

Schema: config

System configuration (schema created; tables managed via application migrations).

No tables defined in this schema in the current SQL file. Configuration tables are managed via application-layer migrations.

Trigger Functions

Two reusable PL/pgSQL trigger functions are defined at the database level and shared across all schemas. Each mutable table references them via CREATE TRIGGER statements. This pattern minimises database objects and ensures consistent behaviour.

public.set_updated_at()

Trigger Function	public.set_updated_at()
Applied To	Shared function
Event	BEFORE UPDATE
Level	FOR EACH ROW
Description	Sets NEW.updated_at = NOW() on any update. Applied to all mutable tables.
Tables Using It	lookup.product_versions → trg_product_versions_updated_at lookup.modules → trg_modules_updated_at lookup.sub_modules → trg_sub_modules_updated_at auth.users → trg_users_updated_at auth.roles → trg_roles_updated_at auth.groups → trg_groups_updated_at company.companies → trg_companies_updated_at company.contacts → trg_contacts_updated_at workflow.ticket_status_graphs → trg_tsg_updated_at workflow.ticket_status_graph_scopes → trg_tsgs_updated_at ticket.tickets → trg_tickets_updated_at ticket.ticket_comments → trg_ticket_comments_updated_at sla.sla_policies → trg_sla_policies_updated_at sla.sla_tickets → trg_sla_tickets_updated_at sla.automation_rules → trg_automation_rules_updated_at kb.kb_folders → trg_kb_folders_updated_at kb.kb_articles → trg_kb_articles_updated_at kb.portal_banners → trg_portal_banners_updated_at notification.notification_templates → trg_notification_templates_updated_at gamification.point_rules → trg_point_rules_updated_at gamification.agent_streaks → trg_agent_streaks_updated_at

ticket.generate_ticket_number()

Trigger Function	ticket.generate_ticket_number()
Applied To	ticket.tickets
Event	BEFORE INSERT
Level	FOR EACH ROW
Description	Auto-generates TKT-YYYYMM-NNNNN ticket numbers using ticket.ticket_number_seq sequence. Only fires when
Tables Using It	ticket.tickets → trg_tickets_number

Custom Types & Enums

Type	Values
ticket.ticket_priority	urgent, high, medium, low
ticket.ticket_source	email, portal, phone
ticket.ticket_status	new, open, in_progress, pending_customer, pending_internal, on_hold, product_roadmap, pending_approval, compliance_review, dual_agent_confirm, resolved, reopened, closed
ticket.ticket_type	bug, enhancement, incident, change_request, query, service_request
notification.notification_channel	email, in_app, sms
notification.delivery_status	sent, failed, bounced, pending
gamification.badge_tier	bronze, silver, gold, platinum
gamification.period_type	daily, weekly, monthly, all_time
auth.revocation_reason	logout, suspicious_reuse, admin_force, role_change
ai.suggestion_type	troubleshooting_steps, similar_tickets
ai.agent_rating	relevant, moderately_relevant, not_relevant
company.deployment_model	cloud, on_premise, hybrid
sla.trigger_event	ticket_created, ticket_updated, time_based